

HiveQL commands

A .Create , drop ,alter table

1. java -version -----check installation & give version
2. hadoop version -----check installation & give version
3. hive -version -----check installation & give version
4. start-dfs.sh -----Starts the HDFS (Hadoop Distributed File System) services.
5. start-yarn.sh -----Starts YARN (Yet Another Resource Negotiator) services.
6. Jps -----Lists all running Java processes on your system.

A typical workflow:

- Run start-dfs.sh → starts storage (HDFS)
- Run start-yarn.sh → starts computation/resources (YARN)
- Run jps → confirm everything is running

7. Hive ----- start hive
8. CREATE DATABASE flightdb;
9. SHOW DATABASES;
10. USE flightdb;
11. SHOW TABLES;
12.
 - CREATE TABLE flights (
 - > flight_id INT,
 - > airline STRING,
 - > departure_delay INT
 - >)
 - > ROW FORMAT DELIMITED
 - > FIELDS TERMINATED BY ','
 - > STORED AS TEXTFILE;

13. open new ubuntu terminal and create a file name as flights.csv

“nano flights.csv” also add some data

1	Indigo	10
2	AirIndia	20
3	SpiceJet	5
4	Vistara	15

14. > LOAD DATA LOCAL INPATH '/home/pratik/flights.csv'

> INTO TABLE flights;

15. SELECT * FROM flights; -----print data

16 . DROP TABLE flights;-----delete table

17. SHOW tables;----confirm delete or not.

18. DO 12,14,15 again.

19. DESCRIBE flights;

20. ALTER TABLE flights

> ADD COLUMNS (destination STRING);----new column added

21. INSERT INTO flights VALUES (5, 'AkasaAir', 12, 'Mumbai');

B .Create external table

1. dfs -mkdir -p /user/pratik/externalflights; ---This creates folder in Hadoop storage

2. dfs -put /home/pratik/flights.csv /user/pratik/externalflights; ----
Put file into HDFS

3. CREATE EXTERNAL TABLE flights_ext (

flight_id INT,

airline STRING,

departure_delay INT

)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ','

STORED AS TEXTFILE

LOCATION '/user/pratik/externalflights';

4. SHOW TABLES;

5. SELECT * FROM flights_ext;

C . JOIN operation :-

1. CREATE TABLE airlines (
 airline STRING,
 country STRING
)

ROW FORMAT DELIMITED

FIELDS TERMINATED BY ','

STORED AS TEXTFILE;

2. nano airlines.csv ----open file

3. Indigo,India

AirIndia,India-----paste some data

4. LOAD DATA LOCAL INPATH '/home/pratik/airlines.csv'

INTO TABLE airlines;

5. SELECT * FROM airlines;-----ALL 5 SIMILAR AS ABOVE STEPS

6. JOIN ➔

- SELECT f.flight_id, f.airline, f.departure_delay, a.country
- FROM flights f
- JOIN airlines a
- ON f.airline = a.airline;

#d. Create index on Flight Information Table

- CREATE INDEX idx_airline
- ON TABLE flights (airline)
- AS 'COMPACT'
- WITH DEFERRED REBUILD;

- ALTER INDEX idx_airline ON flights REBUILD;

#e. Find the average departure delay per day in 2008.

1. nano flights_2008.csv --Create new file for 2008 data
2. 10,Indigo,12,Delhi,2008-01-01
 - i. 11,AirIndia,18,Mumbai,2008-01-01
 - ii. 12,SpiceJet,14,Pune,2008-01-02
 - iii. 13,Vistara,6,Delhi,2008-01-02
3. LOAD DATA LOCAL INPATH '/home/pratik/flights_2008.csv'
 - i. INTO TABLE flights;
4. SELECT * FROM flights;
5. SELECT flight_date,
 - AVG(departure_delay) AS avg_delay
 - FROM flights
 - WHERE flight_date LIKE '2008%'
 - GROUP BY flight_date;